THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DAR ES SALAAM REGION
FORM FOUR - MOCK EXAMINATION - 2026

CHEMISTRY 2A
(ACTUAL PRACTICAL)

CODE:032/2A
TIME:2:30Hours

Thursdat 21st May, 2026 A.M

INSTRUCTIONS

i. This paper consist of two questions. Answer **ALL** questions.

ii. Each question carries twenty five (25) marks making a total of 50 marks.

iii. Cellular phones and any unauthorized materials are **not** allowed in the examination room.

iv. Write your **Examination Number** on every page of your answer booklet.

v. You may use the following constants:
Atomic masses C = 12, O = 16, H = 1, K = 39, Na = 23, N = 14
$1\ dm^3 = 1L = 1000cm^3$

1. You are provided with

1.1 Solution O made by diluting $200cm^3$ of 2M acetic acid (CH_3COOH) to make 2 litres of solution.

1.2 Solution R made by dissolving 1.45 g of impure potassium hydroxide to make 250 cm^3 solution.

1.3 Phenolphthalein indicator (P.O.P)

Procedures

(i) Pipette 25 cm^3 of solution R into conical flask

(ii) Add two drops of P.O.P indicator into conical flask

(iii) Titrate solution O against solution R Obtain three more readings and record your results in a tabular form

Questions

(a) (i) The colour change was from __ to __

(ii) Calculate the average volume of acid used

(b) Find (i) Number of moles of acetic acid used in the reaction

(ii) Number of moles of potassium hydroxide used in the reaction

(c) Calculate the molarity of acid (CH_3COOH)

(d) Calculate the percentage purity of potassium hydroxide

2. As a chemist working at Non Governmental laboratory department, you are required to identify cation and anion present in a salt sample W brought in your office. Carry out the following experiments and record your observation in a tabular form. Appearance, Action of heat on a sample, Action of dil HCl on a sample, Solubility, Action of NH_3 solution on a sample solution, Flame test and addition of $MgSO_4$ on a sample solution followed by warming the content.

CONCLUSION

(a) (i) Cation present is __

Anion present is __

(ii) Chemical formula of a sample is __

Chemical name of a sample is __

(b) Explain the uses of salt sample W in daily life